

Tutorial No. 10

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Problem Statement: Design test cases using appropriate test case design techniques for a login module and explain the role of software maintenance, configuration management, and risk management in long-term software success.

Solution:

Introduction

A login module is a critical component of any software system that ensures secure access to authorized users. Designing effective test cases using structural techniques helps improve reliability, security, and usability. Additionally, long-term software success depends on robust software maintenance, configuration management, and risk management practices.

Test case design for login module

Functional requirements of login module

User should enter valid username and password

System should authenticate credentials

Error message for invalid inputs

Account lock after multiple failed attempts

Password masking for security

2.2 Test case design techniques used

- Equivalence Partitioning (EP)
- Boundary value analysis (BVA)
- Decision table testing
- State transition testing

2.3 Test cases using Boundary value analysis

test case ID	input	boundary condition	Expected
TC06	Username length = min-1	below minimum	error
TC07	Username length = min	Valid boundary	accept
TC08	Username length = max	Valid boundary	accept
TC09	Username length = max+1	Above maximum	error

2.4 Decision table testing

Condition	valid username	valid password	out
Rule 1	yes	yes	log
Rule 2	yes	no	e
Rule 3	no	yes	e
Rule 4	no	no	e

2.5 State transition testing

Current state	input	next state
Idle	Enter credentials	validating
Validating	correct credentials	logged in
Validating	incorrect credentials	error
error	retry	Validating
error (3 attempts)	retry	Account locked

Role of Software Maintenance

Software maintenance ensures the system remains functional and reliable after deployment.

Types of maintenance

Corrective maintenance - fixing bugs and errors

Adaptive maintenance - updating software for new environments

Perfective maintenance - Enhancing features and performance

Preventive maintenance - Improving code quality to avoid future issues.

Importance

Improves system lifespan

Enhances user satisfaction

Keeps system compatible with evolving technology

Reduces long-term costs.

Role of configuration management

Configuration management (CM) controls and tracks changes in software artifacts.

Key activities

Version control (eg. git)

Change management

Enable rollback to stable versions.

Supports team collaboration.

Role of Risk management

Risk management identifies, analyzes, and mitigates potential risks.

Types of risks

Technical risks (system fail failure, bugs)

- Project risks (delay, cost overrun)
- Security risks (data breaches)

Risk management process

- 1.) Risk identification
- 2.) Risk analysis
- 3.) Risk mitigation
- 4.) Risk monitoring

Importance

- Minimizes project failures
- Improves decision-making
- Enhances system security and reliability

6. Conclusion

Effective test case diagram design ensures robust login system by identifying potential errors early. Alongside testing, software maintenance, configuration management, and risk management play a crucial role in ensuring long-term success, scalability, and reliability of software systems.